Rule-table structure of life-like cellular automata: activity, asymmetry, sensitivity, and the propagation–storage factorisation

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ABSTRACT

What structure on the rule space of a cellular-automaton family exposes which rules can support which behaviours? For life-like rules—the two-dimensional family that contains Conway's Game of Life—the standard coordinate is Langton's λ, a single number measuring how much of the rule table fires, long known to be too coarse. We argue for a small, symmetry-organised set of rule-table coordinates and, more pointedly, for what such coordinates can and cannot do: built from cardinalities, they are *necessary but not sufficient* to certify a rule's behaviour, yet they can *exclude* behaviour outright. We split Langton's parameter into its birth and survival contributions, whose difference is a birth–survival asymmetry that separates birth-dominated from survival-dominated rules—a coordinate specific to the outer-totalistic family (whose rule table factors by cell state into a birth set and a survival set), not a re-parameterisation of λ that carries over to general CA. We add the average Boolean-derivative sensitivity of Vichniac and Binder, extended to the life-like family in closed form, whose centre-cell component vanishes exactly on the fully totalistic rules (those whose birth and survival conditions coincide). Combined with an exhaustive enumeration, across all $262{,}144$ rules, of the compact gliders, still lifes and oscillators each rule supports, these coordinates expose a clean structure. Propagation and storage separate along the asymmetry axis: the birth rule is a narrow gate deciding whether anything can move, the survival rule tunes how much structure persists once that gate is open, and—at the compact footprint—the rules supporting gliders are systematically birth-light and survival-rich. One exclusion is exact: no fully totalistic ($B = S$) rule supports a compact (3×3, period ≤ 16) glider, an exhaustively verified combinatorial exclusion that the vanishing centre-cell sensitivity explains, while the same rules oscillate at near-background rate—localising it to translation rather than to in-place recurrence. Two dynamical results frame the whole: rules that support a glider yet rarely or never produce one under random sampling show that capability is *certifiable* from the table but its expression only *estimable* from dynamics; and, conversely, of the uncatalogued rules whose dynamics we screen for a Game-of-Life–like object ecology, those matching it richly are all Conway or HighLife variants differing only on neighbour counts their trajectories never visit—so table difference need not mean behavioural difference. The coordinates organise the family not by certifying capability but by carving the regions where it is possible—and, on the fully totalistic submanifold, the region where it is impossible.

1 Introduction

A cellular automaton (CA) updates a lattice of cells in parallel by a rule that depends only on each cell and a small neighbourhood, yet even tiny rule families produce propagating structures, long transients and universal computation (von Neumann 1966, Wolfram 1983, Berlekamp et al. 1982). This paper concerns a basic organisational question for one such family: given the $262{,}144$ two-dimensional outer-totalistic "life-like" rules, what coordinates on the rule table expose the structure of behavioural capability—which regions support propagation, which support storage, and which exclude a behaviour outright? A life-like rule is fixed by two subsets $B, S \subseteq \{0, \ldots, 8\}$: a dead cell is born if its live-neighbour count lies in B, and a live cell survives if its count lies in S. The fam-

ily contains Conway's Game of Life and its named variants—HighLife, Day & Night, Seeds, 2×2, Replicator, Maze (Gardner 1970, Adamatzky 2010, Eppstein 2010).

The standard coordinate is Langton's λ, the fraction of the rule table that yields a live successor (Langton 1990). Sweeping λ moves through frozen, periodic, complex and chaotic regimes, and the intermediate "edge of chaos" was conjectured to host maximal computational capability. But λ is a single number summarising an 18-bit table: two rules that differ in *which* neighbour counts fire, not just *how many*, collapse onto the same point, and evolved-CA studies found no sharp critical λ (Mitchell et al. 1993).

Necessary, not sufficient. This collapse is not a defect to be engineered away but a structural fact worth stating plainly, because it sets the ceiling on what any cardinality-based coordinate can do. Both λ and the asymmetry δ we introduce below are functions of the cardinalities $(|B|, |S|)$ alone; there are only 10×10 such pairs, so the entire (λ, δ) plane has at most 100 occupied points, each shared by thousands of behaviourally distinct rules. A coordinate this coarse can never *certify* that a rule does something—any value it takes is shared by a crowd. What it can do is the opposite: place a rule in a region, and—in the limiting case—*exclude* a behaviour entirely. Our results are organised around this asymmetry. The asymmetry coordinate δ locates where capability concentrates (a necessary condition); the centre-bit sensitivity μ_{centre} supplies the one exact exclusion, ruling out compact gliders on the fully totalistic submanifold outright.

The coordinates. We add two complementary rule-table coordinates. The first comes from splitting λ into its birth and survival halves, $\lambda_B = |B|/9$ and $\lambda_S = |S|/9$; their difference $\delta = \lambda_B - \lambda_S = (|B| - |S|)/9$ is a birth–survival asymmetry, the antisymmetric companion to Langton's λ, and the coordinate on which our central structural result is read. The split is only defined for outer-totalistic families whose rule table separates by cell state (dead-cell births B vs. live-cell survivals S); in that sense δ is a coordinate for this family, not a re-parameterisation of λ that generalises to CA whose transitions do not factor this way. The second is the average Boolean-derivative *sensitivity* μ—the probability that flipping one input bit of the neighbourhood flips the output—introduced for CA by Vichniac (Vichniac 1990) and used as an a-priori ordering parameter by Binder (Binder 1993). For life-like rules μ admits a closed form that splits into a centre-bit part μ_{centre} and a neighbour part μ_{neigh}, and the centre part vanishes exactly on the diagonal $B = S$. The two new coordinates are independent, not interchangeable: under the birth/survival swap $\tau : (B, S) \mapsto (S, B)$, δ is odd while μ is even, so μ cannot tell a rule from its $B \leftrightarrow S$ mirror and δ can. With λ and the bit-flip complement σ they organise the rule table into a four-sector parity decomposition, closed by one further coordinate δ_{edge} (§2.4).

Findings. Pairing this coordinate system with an exhaustive combinatorial enumeration of every rule's compact-spaceship, still-life and oscillator support, we establish three results. (i) A hierarchical factorisation of capacity, read off the birth–survival asymmetry δ: the birth set is a narrow gate that decides whether anything can propagate, while the survival set tunes how much structure is stored within that gate. The gate structure is robust across seed footprints (the 3×3 propagation-enabling birth sets are a subset of the 4×4 ones, both dominated by small birth sets); at the compact footprint this shows up as a clean fingerprint—the rules supporting compact gliders are systematically birth-light and survival-rich ($\delta < 0$)—which is the paper's central structural finding, and which we show is specific to the compact footprint and dilutes as the window widens. (ii) Three coordinate projections of the rule table—the Binder (λ, μ) plane and its (δ, μ) and (λ, δ) companions—that organise the family non-trivially, in which the 512 fully totalistic rules occupy a sharply identifiable locus. (iii) A total exclusion of $B = S$ rules from compact (3×3, period ≤ 16) gliders—certified by the exhaustive enumeration and explained by the vanishing centre-cell sensitivity $\mu_{\text{centre}} = 0$—which the oscillator enumeration localises to translation rather than to in-place recurrence, and which softens into a widening handicap rather than vanishing at larger footprints. Two dynamical results frame these. First, a population of rules that support a glider combinatorially yet rarely or never express one under random sampling shows that capability is a rule-table property, distinct from what random dynamics reveal (auxiliary dynamical observables are correspondingly reframed as regime descriptors, not capability detectors). Second, and conversely, a table-blind screen of the uncatalogued build-and-hold rules for Game-of-Life–like dynamics returns a richly Life-like ecology only for Conway and HighLife variants—rules whose tables differ from an anchor's solely on neighbour counts their trajectories never visit—so that table difference need not imply behavioural difference. The exact bit-flip duality σ (§2.4) is used only to gauge-fix that auxiliary dynamical sweep (Appendix A).

2 Methods

2.1 Notation and computed quantities

Table 1 collects every rule-table and dynamical quantity used in the paper. The rule-table coordinates are defined in §2.2–§2.4, the combinatorial indices in §2.5, and the single dynamical probe in §2.6. All rule-table quantities are functions of (B, S) alone and are independent of any simulation protocol.

Symbol	Definition	Measures
λ_B, λ_S	$\lvert B\rvert/9,\ \lvert S\rvert/9$	how much of the birth / survival rule fires, as a fraction
λ	$\frac{1}{2}(\lambda_B + \lambda_S) = (\lvert B\rvert + \lvert S\rvert)/18$	Langton's parameter: the overall fraction of the table that fires
δ	$\lambda_B - \lambda_S = (\lvert B\rvert - \lvert S\rvert)/9$	birth-versus-survival imbalance: positive if the rule births more than it preserves, negative if it preserves more than it births; the paper's factorisation axis
μ	avg. over inputs/bit-positions of P[flip flips output]	sensitivity: how often flipping one input cell flips the rule's output
μ_{centre}	$P[n \in B\triangle S],\ n \sim \mathrm{Bin}(8, \frac{1}{2})$	how much a cell's *own* state matters to its fate; zero exactly when $B = S$
μ_{neigh}	binomially-weighted edge density of B, S	how "jagged" the thresholds are—how often firing switches on/off between adjacent neighbour counts
$e(M)$	$\#\{k : \mathbf{1}[k \in M] \neq \mathbf{1}[k{+}1 \in M]\}$	number of on/off switches in a set M as the count rises
δ_{edge}	$(e(B) - e(S))/8$	whether the birth threshold is more or less contiguous than the survival one
Σ	# distinct 3×3 D_4-orbit spaceships, $T \leq 16$	how many compact gliders the rule can support at all
$\#\mathrm{SL}$	# distinct 3×3 D_4-orbit still lifes (T=1)	how many small static patterns the rule can hold fixed
Ω	# distinct 3×3 D_4-orbit oscillators, $2 \leq T \leq 16$	how many small patterns the rule can hold as in-place blinkers
$\mathcal{B}_{\text{prop}}, \mathcal{B}_{\text{osc}}, \mathcal{B}_{\text{sl}}$	birth sets with ≥ 4 companion S giving $\Sigma > 0$ ($\Omega > 0$; $\#\mathrm{SL} > 0$)	which birth rules "open the gate" for propagation / oscillation / storage
$\mathcal{G}^*_{\text{alt}}$	peak off-axis residual spatiotemporal autocorrelation	whether random dynamics actually show a coherent moving structure
gap	$\mu - \langle\text{sens}\rangle_{\text{traj}}$	how much the dynamics under-use the rule's intrinsic sensitivity

Table 1: Notation and computed quantities. The first eight rows are rule-table coordinates (functions of (B, S) alone); Σ, $\#\mathrm{SL}$, Ω count the small moving / static / blinking patterns a rule supports; the last two are the single dynamical probe. $B\triangle S = (B \setminus S) \cup (S \setminus B)$ is the set of counts where birth and survival disagree; τ and σ are the swap and complement symmetries of §2.4; D_4 is the eight-fold symmetry group of the square.

2.2 Rule-space coordinates: λ, δ

Each rule splits its look-up table into a birth component B on dead cells and a survival component S on live cells, so we define $\lambda_B = \frac{1}{9}\sum_k \mathbf{1}[k \in B]$, $\lambda_S = \frac{1}{9}\sum_k \mathbf{1}[k \in S]$, the Langton parameter $\lambda = \frac{1}{2}(\lambda_B + \lambda_S)$, and its antisymmetric companion

$$\delta = \lambda_B - \lambda_S = \frac{|B| - |S|}{9}. \tag{1}$$

The pair (λ, δ) is a bijective reparameterisation of (λ_B, λ_S).

$\delta = 0$ versus $B = S$. These loci are easily conflated but combinatorially distinct, and the distinction is exactly the necessary/sufficient one. $\delta = 0$ is the equal-cardinality submanifold $|B| = |S|$, containing $\sum_k \binom{9}{k}^2 = \binom{18}{9} = 48{,}620$

rules; the strictly diagonal $B = S$ (the fully totalistic rules) is far smaller, only 512. This $B = S$ set is not new: it is the classical *totalistic* Life-like subclass, whose members are those for which birth and survival coincide (up to the usual centre-counting convention); what is new is its characterisation as the exact zero set of the centre-cell sensitivity below, and the compact-glider exclusion we prove on it. Equivalently, writing the unweighted set distance $|B\triangle S|$ (the number of counts in exactly one of B, S), $B = S$ is $|B\triangle S| = 0$ while $\delta = 0$ is only $|B| = |S|$, so $\{B = S\} \subsetneq \{\delta = 0\}$. The coordinate that isolates $B = S$ exactly is the centre-bit sensitivity introduced next, of which $|B\triangle S|$ is the unweighted shadow: we show in §3.1 that $B = S$ is precisely the $\mu_{\text{centre}} = 0$ locus.

2.3 Sensitivity coordinate μ (Boolean derivative)

The (λ, δ) coordinates summarise activity but say nothing about how strongly a rule responds to single-bit perturbations of its input. The standard summary is the average Boolean derivative (Vichniac 1990, Binder 1993): for a rule with N input bits and binary output o,

$$\mu = \frac{1}{N\,2^N} \sum_{x\in\{0,1\}^N} \sum_{i=1}^{N} \big[o(x \oplus e_i) \neq o(x)\big], \tag{2}$$

the probability that a single-bit flip flips the output; it is closely related to Wuensche's Z parameter (Wuensche 1999). For an outer-totalistic life-like rule on the Moore neighbourhood the $N = 9$ inputs split as one centre bit and 8 neighbour bits, and the table depends on inputs only through (c, n) with c the centre value and n the neighbour count. With $o(c, n) = \mathbf{1}[c{=}0]\,\mathbf{1}[n \in B] + \mathbf{1}[c{=}1]\,\mathbf{1}[n \in S]$, the centre and neighbour contributions have closed forms

$$\mu_{\text{centre}} = \frac{1}{2^8} \sum_{n=0}^{8} \binom{8}{n} \mathbf{1}[n \in B\triangle S], \tag{3}$$

$$\mu_{\text{neigh}} = \frac{1}{2^8} \sum_{k=0}^{7} \binom{7}{k} \big(\mathbf{1}[k \in B\triangle(B{-}1)] + \mathbf{1}[k \in S\triangle(S{-}1)]\big), \tag{4}$$

with $\mu = \frac{1}{9}(\mu_{\text{centre}} + 8\,\mu_{\text{neigh}})$. The notation needs a word. $B\triangle S$ is the *symmetric difference* of the two sets: the neighbour counts at which birth and survival *disagree*—a count belongs to $B\triangle S$ when it triggers exactly one of the two, birth or survival, but not both and not neither. For Conway ($B = \{3\}$, $S = \{2, 3\}$) the two sets share the count 3 and differ only at 2, so $B\triangle S = \{2\}$: a cell's own state matters only when it has exactly two live neighbours. The companion term $B\triangle(B{-}1)$ applies the same "disagree" operation to a set and its shift-by-one, which simply marks the *edges* of B—the counts where membership switches on or off as the neighbour total climbs by one (the symbol $\oplus$ is this exclusive-or of the two adjacent memberships). Three consequences follow.

1. $\mu_{\text{centre}} = 0$ **is the** $B = S$ **submanifold.** By the formula $\mu_{\text{centre}} = 0$ iff $B\triangle S = \varnothing$, i.e. $B = S$. So μ_{centre} is exactly the binomially-weighted set distance from totalistic, and the 512 fully totalistic rules are characterised a priori: they cannot respond differently to the centre cell's own state at any count.
2. μ_{neigh} **counts birth/survival edges**; alternating sets such as $\{1, 3, 5, 7\}$ (Replicator) maximise it.
3. μ **is independent of** λ: one measures the fraction of table entries that fire, the other the fraction of input-bit flips that change the output. Together they form a Binder (λ, μ) plane (§3.1).

2.4 A symmetry-organised coordinate system

The rule table carries two natural involutions,

$$\tau : (B, S) \mapsto (S, B), \qquad \sigma : (B, S) \mapsto \big(\overline{\text{rev}(S)}, \overline{\text{rev}(B)}\big), \tag{5}$$

where $\text{rev}(M) = \{8 - k : k \in M\}$ and the bar is the complement in $\{0, \ldots, 8\}$. τ swaps birth and survival; σ is the bit-flip duality, in the sense that if $g = \sigma(f)$ then $g(\neg x) = \neg f(x)$ for every configuration. Both square to the identity, and the rule space falls into the four parity sectors of (τ, σ). The Boolean-derivative formalism supplies one further coordinate that completes the decomposition, the edge imbalance $\delta_{\text{edge}} = (e(B) - e(S))/8$.

Parity table. We verified the action of τ and σ on every coordinate by direct evaluation on all 262,144 rules (max residue 1.1×10^{-16}): λ is (τ-even, σ-odd; $\lambda \mapsto 1 - \lambda$); δ is (τ-odd, σ-even; δ invariant under σ); $\mu_{\text{centre}}, \mu_{\text{neigh}}, \mu$ are (τ-even, σ-even); and δ_{edge} is (τ-odd, σ-odd). The load-bearing observation is that λ and μ are both τ-even, so neither can distinguish a rule from its $B \leftrightarrow S$ mirror; δ is the unique τ-odd coordinate among the activity/sensitivity axes, and is therefore the natural birth-versus-survival scalar. Since σ acts as $(\lambda, \delta, \mu) \mapsto (1 - \lambda, \delta, \mu)$, self-dual rules need not have $\delta = 0$—consistent with Day & Night ($\delta = -\frac{1}{9}$) being self-dual. Pearson correlations across the full rule space are exactly zero between every pair of $\{\lambda, \delta, \mu, \delta_{\text{edge}}\}$, as required by the orthogonality of parity sectors under the uniform measure.

Roles of the coordinates. Each answers a distinct question the others structurally cannot, and the necessary/sufficient distinction runs through all four. λ places a rule on the frozen→chaotic spectrum but collapses rules of equal total firing fraction, so it is a necessary entry point, never sufficient. δ is the swap-odd factorisation axis: $\Sigma > 0$ rules are birth-light and survival-rich ($\delta \approx -0.22$), the rule-table fingerprint of the B-gate / S-knob structure (§3.4); without δ the claim cannot even be stated. μ_{centre} is the one coordinate that excludes rather than merely locates: $\mu_{\text{centre}} = 0$ iff $B = S$, turning the empirical $B = S$ exclusion into a one-line characterisation (§3.2). μ_{neigh} measures threshold jaggedness but is swap-even and carries no birth-versus-survival information; δ_{edge} is its swap-odd companion, a secondary, mechanism-consistent preference for contiguous birth thresholds (§3.4).

2.5 Combinatorial enumeration: Σ, $\#\mathrm{SL}$, Ω

The following are functions of the rule table alone and independent of any simulation protocol. For each rule we enumerate every 3×3 binary pattern (reduced to its D_4 orbit representative) on a quiescent zero-padded background, evolve it to $T_{\max} = 16$, and classify the orbit by its return behaviour: a nonzero-offset translated return is a *spaceship* (Σ counts distinct orbit-representative spaceships, with period and velocity set); a same-position return at period $2 \le T \le 16$ is an *oscillator* (Ω); a fixed pattern (T=1) is a *still life* ($\#\mathrm{SL}$). Σ asks "can the rule support a compact glider at all," independent of whether a random trajectory ever exhibits one. A cheap pre-screen exploits that translated and stationary copies both preserve the seed's live-cell count exactly: monitoring the running sum rejects the great majority of rules within a few steps, leaving the small fraction that warrant the full check.

2.6 Dynamical probe and coherent-propagation index $\mathcal{G}^*$

To identify rules whose combinatorial propagation is or is not expressed by random dynamics we run a single fixed sweep. For each rule we draw 5 shared Bernoulli ICs at density $\rho_0 = 0.30$ on a 128×128 torus, together with their bit-flipped partners at density 0.70 (the symmetric-IC ensemble; see Appendix A), evolve 600 steps, and measure on the 200-frame tail a residual spatiotemporal autocorrelation $\mathcal{G}^*_{\text{alt}}$: a coherent structure of period T and velocity (v_x, v_y) produces an off-axis peak at $(\Delta t, \Delta x, \Delta y) = (T, -Tv_x, -Tv_y)$, validated by an injection test in which an isolated Conway glider peaks at $(\Delta t{=}4, \Delta x{=}-1, \Delta y{=}-1)$. We use $\mathcal{G}^*_{\text{alt}}$ only to flag rules with no measurable coherent propagation; the remaining dynamical observables are regime descriptors, deferred to Appendix A.

3 Results

3.1 The rule-table phase diagrams (λ, μ), (δ, μ), (λ, δ)

The sensitivity coordinate places every rule on a second axis complementing λ (Figure 1). The cloud of all 262,144 rules forms a structured band, and the $B = S$ submanifold ($\mu_{\text{centre}} = 0$) traces a sharply identifiable locus on which all 512 fully totalistic rules lie. Replicator sits at the top with $\mu = 0.889$—centre-insensitive but maximally neighbour-sensitive—showing $B = S$ is not synonymous with low sensitivity, only with centre-flip insensitivity. The 424 rules with $\Sigma > 0$ cluster off that locus at moderate λ and $\mu \in [0.25, 0.60]$; Conway ($\mu = 0.35$) and HighLife (0.46) are $\Sigma > 0$ and lie in the cluster, while the storage-heavy Class IV anchor Day & Night lies at the same moderate μ (0.565) yet is itself $\Sigma = 0$ (Figure 5)—a first illustration that the moderate-(λ, μ) region is necessary for compact propagation but does not certify it (Wolfram 1984). (2×2, another Class IV anchor with $\Sigma = 0$, sits just above the band at $\mu = 0.639$; both rules are complex through engineered rather than compact-glider dynamics, a point we return to in §4.2.) The middle panel shows the same rules in (δ, μ), where the $\Sigma > 0$ cluster sits at $\delta \approx -0.2$ (the birth-light/survival-rich offset of §3.4) and the $B = S$ rules collapse to the line $\delta = 0$; the right panel shows the (λ, δ) projection, where the $\Sigma > 0$ rules form a diamond near $(0.4, -0.2)$. We do not claim the planes as a sharp classifier: the $\Sigma > 0$ band is contained in, but does not exhaust, the moderate region—necessary, not sufficient.

3.2 Combinatorial spaceship support and the $B = S$ exclusion

Of the 262,144 rules, $\Sigma > 0$ holds for 424 (0.16%); these cluster in the (λ, δ) diamond near $(0.4, -0.2)$, overlapping Conway. They support 876 distinct orbit-representative spaceships, of which 560 are diagonal and 316 orthogonal, with no 3×3 knight-ship; the period distribution peaks at $T = 4$. Direction is a clean per-rule property: of the 424, 300 support only diagonal ships and 124 only orthogonal, none both at the 3×3 footprint. Only Conway and HighLife appear among the project's canonical anchors, leaving many previously uncatalogued rules that support a compact glider—though, as §4.2 shows, those with genuinely Conway-like bulk dynamics turn out to be anchor variants rather than new rules (Figure 2).

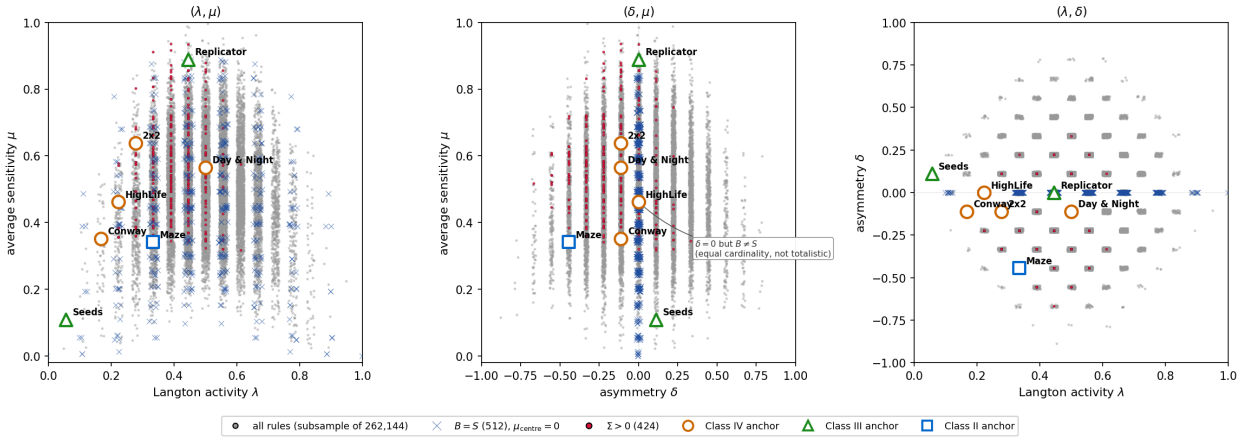


Figure 1: Three coordinate projections of all 262,144 rules (grey subsample), the 512 fully totalistic $B = S$ rules (royal-blue X, $\mu_{\text{centre}} = 0$), and the 424 rules with $\Sigma > 0$ (crimson). Anchors are outlined markers coloured by Wolfram class. (λ, μ): $B = S$ rules trace the $\mu_{\text{centre}} = 0$ locus, $\Sigma > 0$ rules sit off it. (δ, μ): $B = S$ rules collapse to $\delta = 0$ while $\Sigma > 0$ rules cluster at $\delta \approx -0.2$. (λ, δ): $\Sigma > 0$ rules form a diamond near $(0.4, -0.2)$. HighLife and Replicator lie on $\delta = 0$ by equal cardinality without being totalistic, illustrating the $\delta = 0$ vs. $B = S$ distinction of §2.2.

The $B = S$ exclusion and its footprint dependence. The 512 rules with $B = S$ (equivalently $\mu_{\text{centre}} = 0$) contain *zero* rules with $\Sigma > 0$ at any $T \leq 16$, against the bulk rate of 0.16%. This is the sharp form of "exclude, not certify": a single cardinality-blind coordinate, μ_{centre}, rules out compact propagation outright on its zero set. The exclusion is specific to the compact footprint. Enumerating larger seeds exhaustively over all $B = S$ rules, 36 of the 256 vacuum-stable ones support a $4{\times}4$ spaceship and 40 a $5{\times}5$ one (against 0 at $3{\times}3$), so $\mu_{\text{centre}} = 0$ closes the *compact* propagation channel, not propagation at all scales. The recovery is bounded, not monotone toward parity: the count saturates near 15% of vacuum-stable $B = S$ rules, and *build-and-hold*—a glider together with static storage—remains rare, holding for 4 rules at $4{\times}4$ and 8 at $5{\times}5$ (counting still lifes; every glider-bearing rule oscillates, so oscillation is not an informative storage test here). Moreover the gap to Conway *widens* with footprint: the still-life count of the richest such $B = S$ rule trails Conway's by $6.3\times$ at $4{\times}4$ and $9.2\times$ at $5{\times}5$. A rule forced onto the diagonal tunes propagation and storage with a single set and cannot exploit a larger window on both axes at once, as a rule with independent B and S can; the strict $3{\times}3$ exclusion therefore softens into a widening handicap rather than vanishing. The combinatorial recovery does not, however, bring behavioural complexity with it. Running the richest build-and-hold candidates—the rules that combine a larger-footprint glider with the most still lifes—from random initial conditions across a range of densities, and inspecting the trajectories directly, we find chaotic (Class III-like) bulk dynamics: the field boils to a disordered steady state without the enumerated glider visibly emerging or persistent objects surviving. Within the footprints examined ($\leq 5{\times}5$) and the random-initial-condition sampling used, no fully totalistic rule displays Game-of-Life–like (Class IV) behaviour; the diagonal gains combinatorial glider capability at larger scales but not expressed complexity. We give the mechanism in §4.1. (The $4{\times}4$ enumeration is exhaustive over all 262,144 rules—8,206 support a spaceship, and the 424 compact-glider rules are contained in this set exactly; the $5{\times}5$ enumeration is exhaustive over $B = S$ but sampled elsewhere.)

3.3 Glider-capable but dynamically silent rules

Cross-referencing Σ against $\mathcal{G}^*_{\text{alt}}$ isolates a sharp population that makes the capability/expression gap concrete. We find 33 rules that combinatorially support a $3{\times}3$ glider yet produce no coherent propagation from random initial conditions across a deliberately wide search—13 densities $\rho_0 \in [0.05, 0.90]$, 25 seeds each, runs of 10^3 steps (325 trials per rule). The asymmetry between the two claims is the point. That the glider *exists* is a construction, not a sample: enumerated from the table and verified by injecting it onto a quiescent grid, where it propagates steadily. That random dynamics *do not* produce it is, by contrast, only ever an estimate over sampled initial conditions; we state it as a measure, not an absolute—the glider's basin is negligible under random sampling, undetected in those 325 trials. Capability is certifiable from the table; expression is only estimable from dynamics. (A further five rules silent at $\rho_0 = 0.30$ express intermittently at other densities; they are rare-basin, not silent, and are excluded from the 33.)

The obstruction is not a missing background. Every $\Sigma > 0$ rule has a stable dead vacuum by construction—a glider is enumerated on a quiescent background, so a rule whose vacuum ignited could not register $\Sigma > 0$—and indeed all 33 have $0 \notin B$. The glider genuinely lives on empty space; the random-IC trajectory simply never assembles it

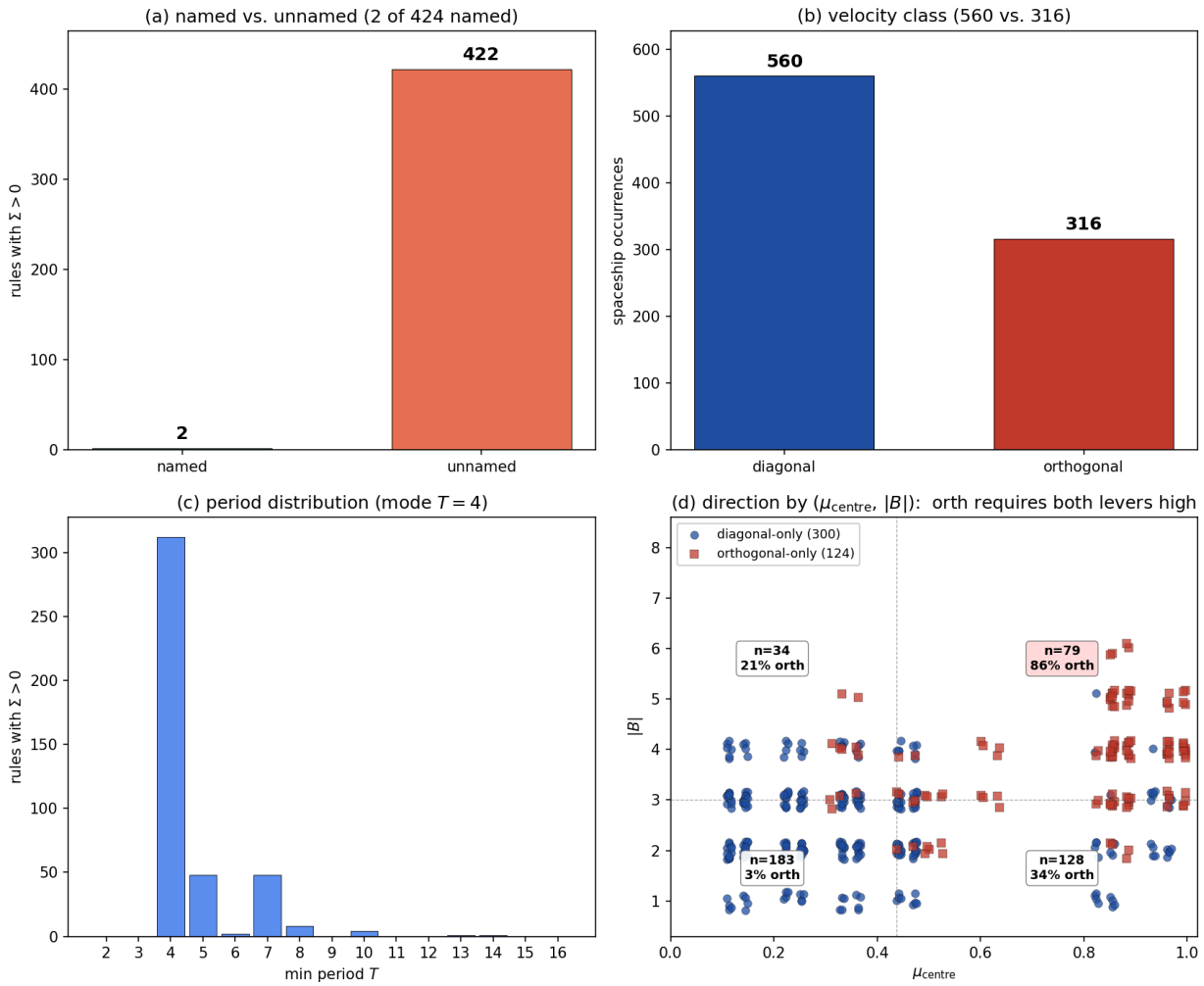


Figure 2: Combinatorial characterisation of the 424 rules with $\Sigma > 0$. (a) Named vs. unnamed in the canonical anchor list (2 of 424). (b) Velocity class at the per-ship level (560 diagonal vs. 316 orthogonal). (c) Period distribution, mode $T = 4$. (d) Direction in the $(\mu_{\text{centre}}, |B|)$ plane: orthogonal motion requires high μ_{centre} *and* large $|B|$ jointly (the joint corner is 86% orthogonal, either lever alone only 21–34%, the unfavoured corner 3%); diagonal is the default elsewhere (§4.1).

there. Measuring the trajectories shows two opposite failures: 22 of the 33 *freeze* (the field settles to near-vacuum, observed tail densities $\bar{\rho} \approx 0.005$–0.012) and 11 *flood* (the field saturates, $\bar{\rho} \approx 0.58$–0.99), with none in the moderate band $[0.10, 0.56]$ that expressing rules—Conway among them—occupy. On the silent population the split is read by a single survival bit: every flood rule has $8 \in S$ (saturated regions persist; $11/11$), while the freeze rules have restrictive survival ($\overline{|S|} = 4.5$, two-thirds with loneliness death $0, 1 \notin S$). This is descriptive of the 33, not a general law: on a uniform 3,000-rule sample from the full 262,144, ranking rules by activity λ alone recovers the flood-versus-freeze (dense-versus-sparse settling) split in the large majority of pairwise comparisons—about 86%—while no simple birth/survival set relation predicts the split out of sample. The silent rules are those whose glider-bearing birth set pairs with a survival rule that drives the trajectory out of the viable band, too dense or too sparse for the glider to persist on clean background. Which regime a rule falls into is determined by the table but not reducible to a closed-form predictor of it: a mean-field density map reproduces the gross density ordering across rules (rank correlation 0.77 with measured $\bar{\rho}$, against 0.34 for the best single-bit feature) yet mis-places most silent rules at a moderate interior fixed point ($\rho \approx 0.4$) the spatially-fragmenting dynamics never reach. The regime must be simulated—the dynamical counterpart of "certifiable versus estimable."

A controlled witness fixes capability and isolates the survival rule. The family $B = \{3\}$ contains rules that share a common supported diagonal glider, Conway among them. Holding the glider fixed and varying only S, some express it from random ICs and others do not: Conway's $S = \{2, 3\}$ does, while $B\{3\}/S\{2, 4, 5, 8\}$—the same birth set, the same supported glider—does not. Expression is thus not a function of capability alone. Propagation capacity is set by B; whether random dynamics express it depends on S (§4.2).

3.4 Hierarchical B/S factorisation of propagation and storage

Define the propagation-enabling birth sets $\mathcal{B}_{\text{prop}} = \{B : |\{S : \Sigma(B, S) > 0\}| \geq 4\}$. Only 24 of 512 birth sets (4.7%) qualify, dominated by $B \supseteq \{3\}$ variants; outside $\mathcal{B}_{\text{prop}}$ propagation capacity is essentially absent. Within the enabled slice (12,288 rules) the Spearman correlation between still-life count and $|S|$ is 0.555; outside the slice it drops to 0.135 (Figure 4). The factorisation is hierarchical, not symmetric: B gates whether any structure is possible, S tunes storage richness once the gate is open.

In-place recurrence forms one storage axis. Enumerating in-place oscillators by the same protocol, 57,270 rules (21.85%) have $\Omega > 0$; all 424 $\Sigma > 0$ rules are among them, 56,846 oscillate without any spaceship, and none support a spaceship without also oscillating. Oscillation is birth-gated through a strictly wider gate: $|\mathcal{B}_{\text{osc}}| = 128$ against $|\mathcal{B}_{\text{prop}}| = 24$, and storage (any still life) through a wider one still, $|\mathcal{B}_{\text{sl}}| = 176$; the three form a strict three-level nesting $\mathcal{B}_{\text{prop}} \subset \mathcal{B}_{\text{osc}} \subset \mathcal{B}_{\text{sl}}$ ($24 \subset 128 \subset 176$ of 512; Figure 4a). Storage is thus wider than oscillation—modestly, not vastly—and propagation is the narrowest gate. We find Spearman $\rho(\Omega, \#\text{SL}) = +0.825$ against $\rho(\Omega, \Sigma) = +0.080$; the informative quantity is the contrast, not the magnitude (still lifes and oscillators are the same enumeration at period 1 and ≥ 2, so a strong correlation is expected). The relevant partition is not static-versus-moving but translating-versus-non-translating: in-place recurrence at any period forms a single S-tuned storage axis, and translation is orthogonal to it for a structural reason—it needs the centre-bit asymmetry $\mu_{\text{centre}} > 0$ that the in-place objects do not. The $B = S$ submanifold confirms this directly: $B = S$ rules support oscillators at $96/512 = 18.75\%$ (within reach of the 21.85% full-space rate, all at period $T = 2$) while supporting $0/512$ compact spaceships. $\mu_{\text{centre}} = 0$ closes the translation channel specifically and leaves the in-place storage channel essentially intact (Figure 4; mechanism in §4.1).

δ as the factorisation axis, δ_{edge} as a secondary one. Because the factorisation is intrinsically a birth-versus-survival statement, the τ-odd coordinate δ is its natural projection. Among the 424 $\Sigma > 0$ rules the δ distribution is sharply skewed to $\delta < 0$ (median -0.222, 340 of 424 below zero), whereas the $\Sigma = 0$ bulk is symmetric about zero by the τ-symmetry of rule space (Figure 3). This $\delta \approx -0.22$ is, for the asymmetry axis, the analogue of Langton's critical $\lambda \approx 0.25$ for the activity axis: a single scalar value where capability concentrates. But the paper's framing applies to it in full—both λ and δ *locate* without *certifying*. The same $\delta \approx -0.22$ slice is shared by thousands of $\Sigma = 0$ rules, so landing on it is necessary for the propagation–storage structure but never sufficient for a glider; the value marks a region, not a guarantee. This fingerprint is specific to the compact footprint: among the 8,206 rules that support a 4×4 spaceship the δ skew washes out (median 0, rank-biserial -0.12 against ≈ -0.5 at 3×3), because a wider window admits birth-heavier orthogonal shippers. What survives footprint widening is the *gate* structure, not the δ sign: the 24 birth sets of $\mathcal{B}^{3\times3}_{\text{prop}}$ are contained in the 64 of $\mathcal{B}^{4\times4}_{\text{prop}}$ exactly, and both remain dominated by small birth sets (mean $|B| = 3.5$ at 4×4, against a uniform peak at $|B| = 5$). The birth-light asymmetry is thus what *compact* propagation costs, not a universal law; the narrow gate of small birth sets is the footprint-robust core. The τ-odd, σ-odd coordinate δ_{edge}, orthogonal to (λ, δ, μ) on the full space, carries a genuine but secondary signal: $\Sigma > 0$ rules favour a birth set that is both small ($\delta < 0$) and contiguous ($\delta_{\text{edge}} < 0$). Because we enumerate the entire rule population rather than a sample, we read these as effect sizes, not significance: conditional on (λ, δ, μ) the δ_{edge} logistic coefficient retains about one quarter the magnitude of δ's ($\beta = -0.21$ vs. -0.78), and stratifying by δ leaves about 70% of the marginal shift—a contiguity preference on top of small cardinality, not a co-equal second axis. The four parity sectors $(\lambda, \delta, \mu, \delta_{\text{edge}})$ together form a complete Boolean-derivative parametrisation; the (odd, odd) sector is mechanism-consistent but quantitatively subordinate to δ.

3.5 Capability is a rule-table property, not a dynamical one

The silent population shows by example that combinatorial capability need not be expressed by random dynamics. We confirm this with the intrinsic-versus-realised sensitivity gap $\text{gap}(f) = \mu(f) - \langle \text{sens} \rangle$, where $\langle \text{sens} \rangle$ averages the Boolean sensitivity over visited rather than uniform configurations. The gap separates the Wolfram-class regimes cleanly (positive for the propagating Class IV anchors Conway and HighLife, negative for the Class III chaos anchors Seeds and Replicator). But asked to separate $\Sigma > 0$ from $\Sigma = 0$ across the enumerated set, the median gaps differ by only -0.34 vs. -0.36, indistinguishable against the intra-class spread. The gap is a clean regime descriptor but not a detector of compact-glider support—a direct measurement of the obstruction the silent population illustrates. Capability

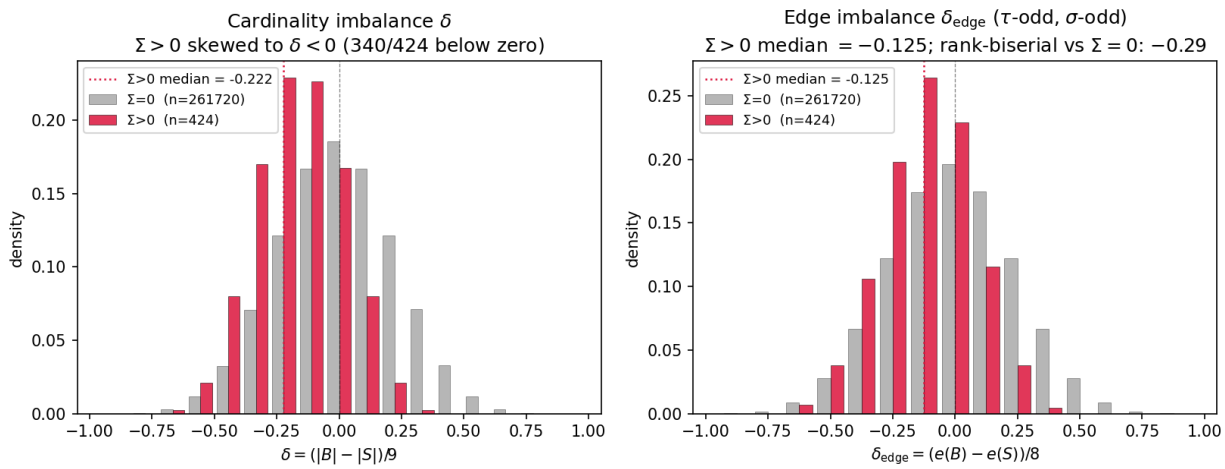


Figure 3: Histograms of the two τ-odd coordinates over all 262,144 rules. Grey: $\Sigma = 0$ bulk; crimson: $\Sigma > 0$ ($n = 424$). Left: δ, symmetric about zero on the bulk (forced by τ-symmetry) but sharply skewed to $\delta < 0$ for $\Sigma > 0$ (median -0.222). Right: δ_{edge}, a similar but partly inherited marginal shift (median -0.125); conditional on (λ, δ, μ) it retains about a quarter of δ's magnitude.

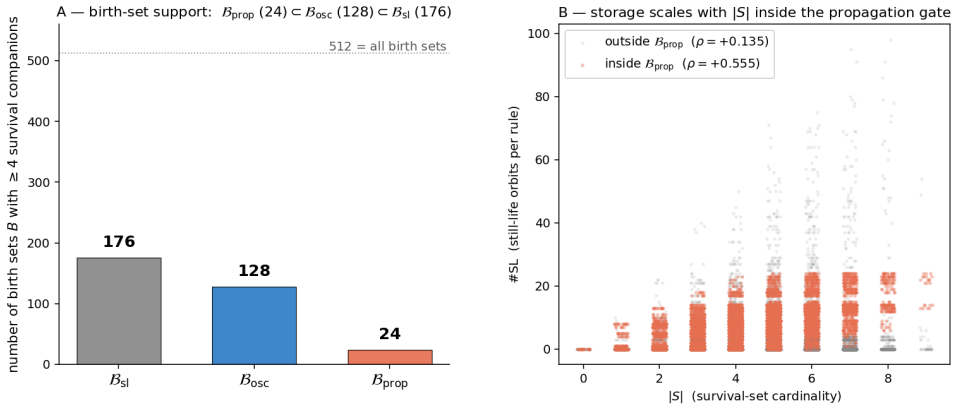


Figure 4: Hierarchical B/S factorisation. (a) Birth-set support of each capability at threshold $k \geq 4$ companions (a birth set counts as supporting a structure iff at least 4 of the 512 possible survival companions S yield it): $|\mathcal{B}_{\text{sl}}| = 176$ (still lifes) $\supset |\mathcal{B}_{\text{osc}}| = 128$ (oscillators) $\supset |\mathcal{B}_{\text{prop}}| = 24$ (spaceships), nested strictly within the 512 birth sets—propagation is the narrowest gate, oscillation wider, storage widest. At the looser $k \geq 1$ threshold $|\mathcal{B}_{\text{prop}}| = 26$ (two borderline birth sets with only two companions each drop out at $k \geq 4$); the storage and oscillation counts are unchanged. (b) #SL against survival-set cardinality $|S|$, split by $B \in \mathcal{B}_{\text{prop}}$ (orange) vs. outside (grey): inside the propagation gate storage tunes with $|S|$ (Spearman $\rho = 0.555$), outside it does not ($\rho = 0.135$).

is certifiable from the table; the dynamical observables of the sweep track regime, not capability, and are reported as such.

4 Discussion

4.1 The mechanism of the $B = S$ exclusion

A compact spaceship needs a leading edge that creates state ahead (a B-condition) and a trailing edge that clears state behind (an $\overline{S}$-condition), on overlapping neighbour-count regions. $\mu_{\text{centre}} = 0$ means the rule is indifferent to a cell's own state at every count, so it cannot resolve the two geometrically-distinct "ahead" and "behind" events—a dead cell becoming live and a live cell dying for the same neighbour configuration—into different outputs. This becomes decisive

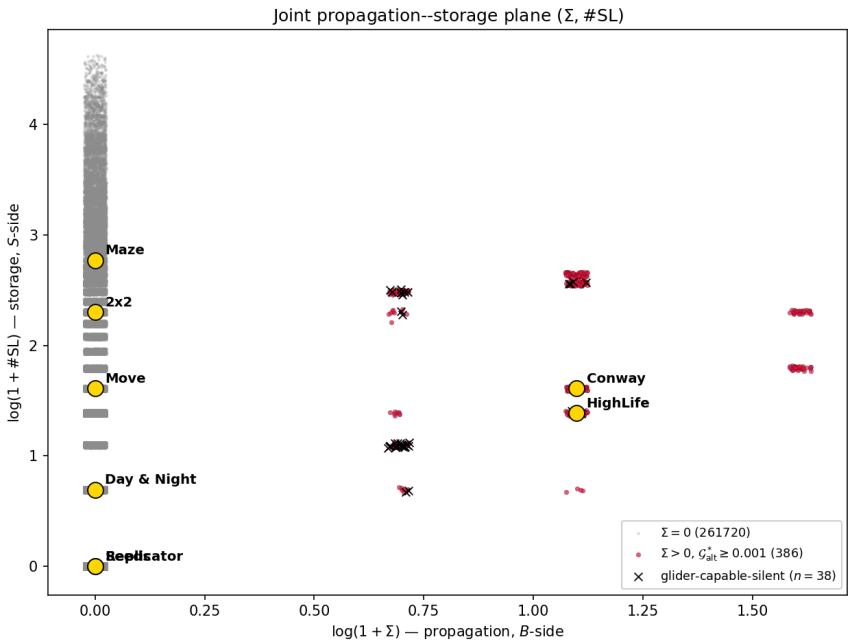


Figure 5: Joint propagation–storage plane $(\Sigma, \#\mathrm{SL})$ in $\log(1+x)$ coordinates. Conway and HighLife sit in the upper-right corner (both gliders and rich storage). The $\Sigma = 0$ column supports no *compact* glider and contains the Class IV anchors 2×2, Day & Night and Move—whose complexity rests on engineered, large-footprint propagation this enumeration does not capture (§4.2)—while Seeds and Replicator sit at the origin. The 38 marked "glider-capable-silent" points are the initial silent population from the main sweep, subsequently resolved as 33 robustly silent and 5 rare-basin rules (§3.3); this 38 is distinct from the 38-rule pre-floor count of the census screen in §4.2, which the two sets happen to share by coincidence.

at the compact footprint for a geometric reason. A 3×3 pattern is exactly the Moore neighbourhood of a single cell: every cell of the object lies within the neighbourhood of every other, so the object fits entirely inside the field of view of one cell in one update. Its leading and trailing edges are therefore never independently addressable—no cell can respond to one without seeing the other—so a rule that cannot distinguish a cell's own state ($\mu_{\text{centre}} = 0$) is forced to treat the "create ahead" and "clear behind" events identically, and the compact channel closes. A 4×4 or larger object breaks this containment: its extremal cells fall outside one another's neighbourhoods, so leading and trailing edges can be driven by disjoint local configurations that $\mu_{\text{centre}} = 0$ no longer collapses. This is precisely why the exclusion is exact at 3×3 and softens as the footprint grows—the single-neighbourhood confinement that makes "compact" meaningful is exactly the condition under which centre-blindness is fatal. The exclusion itself is an exhaustive combinatorial fact (no $B = S$ rule registers a compact spaceship); the centre-cell account here is our explanation of why, tied to a single coordinate. The oscillator results localise it exactly: $\mu_{\text{centre}} = 0$ closes the *translation* channel specifically, because translation needs the leading/trailing asymmetry while in-place recurrence does not; $B = S$ rules oscillate at near-background rate and are excluded only from compact spaceships. This mechanism is heuristic, not yet a theorem (§4.3): the exclusion is proven by enumeration, but a proof *from* the centre-cell argument remains open.

The same centre-bit coordinate also grades the *direction* of translation at the compact footprint (Figure 2d): among 3×3 spaceships orthogonal motion is confined to rules simultaneously high in μ_{centre} and large in $|B|$ (the joint corner is 86% orthogonal, either lever alone only 21–34%), with diagonal the default elsewhere. The two coordinates act roughly additively and neither is reducible to the other. This diagonal dominance is itself compact-specific and inverts with footprint: at 4×4 orthogonal spaceships outnumber diagonal ones by roughly seven to one, the reverse of the 3×3 ratio. The geometric reading—that axial motion needs a sharper, denser leading edge than diagonal motion, and that the extra row of a wider window supplies exactly the structure orthogonal motion requires—we offer only as a hypothesis.

For storage the binding constraint is a B-condition—a still life persists only if its surrounding dead cells have counts outside B—while which interior configurations survive is set by S. This is why $\#\mathrm{SL}$ correlates with $|S|$ only once one conditions on $B \in \mathcal{B}_{\text{prop}}$, closing the factorisation: B gates, S tunes.

4.2 Conway's specialness

Conway sits at the intersection of three conditions, the first two combinatorial and the third dynamical: $B = \{3\} \in \mathcal{B}_{\text{prop}}$ admits the glider; $S = \{2, 3\}$ supports rich still lifes; and the random-IC attractor at $\rho_0 = 0.30$ routes trajectories into a regime that visits the glider on clean background. Rules in the $B = \{3\}$ family share a common supported glider—the first two conditions hold for every one of them—but only a survival rule that keeps the trajectory in the moderate-density band, neither flooding nor freezing, lets random dynamics express it. Conway's S is privileged not because it enables the glider but because it places the dynamics where the glider can appear.

We can ask the converse question directly: among the rules that support both a glider and a still life but are absent from the standard catalogue, are any *dynamically* Conway-like—not merely capable, but producing a Game-of-Life–style ecology of blinkers, blocks and other small objects from random starts? Of the 424 rules with $\Sigma > 0$, the 2 named anchors and 2 further rules with no still life ($\#\mathrm{SL} = 0$) are set aside, leaving 420 uncatalogued build-and-hold rules. We screen these on *behavioural* criteria alone—a stabilising tail density (met by 314 of the 420 at some density), followed by a direct census requiring that canonical Game-of-Life still lifes and oscillators dominate the small live structures, plus a richness floor of at least one canonical still life (block, beehive, loaf, boat or tub) present in the tail, so that a rule producing only sparse blinkers—which are canonical oscillators and would trivially satisfy the raw dominance test—does not qualify. All criteria are applied *without* reference to the rule table, so that table similarity is a finding rather than a filter. Without the richness floor 38 rules pass; with it, 24 remain, and the 14 excluded are precisely the blinker-only degenerates whose census is Life-like in the letter but not in the spirit. Only then do we compare the 24 rich-ecology rules' tables to the anchors: *all* 24 agree with Conway or HighLife on every visited $(\text{centre}, \text{count})$ pair (15 Conway, 9 HighLife), differing only at neighbour counts 7 and 8, which require locally saturated regions and do not fire at the tested density. So every uncatalogued rule that both stabilises and grows a rich Life-like object ecology from random starts is, on its visited configurations, a Conway or HighLife variant—the pool contains no rule whose table is distinct from both anchors yet reproduces their small-object zoo. This is the converse of the paper's leading theme—capability resides in the table but need not be expressed—made concrete on the expression side: two rules can differ wherever their tables are defined yet be dynamically indistinguishable, because behaviour is fixed by the table only on the configurations a trajectory visits. We stress the scope: the screen identifies rules Game-of-Life–*like* in their small-object ecology, not the complete set of dynamically interesting rules—a rule with a rich but non-Life-like object zoo would pass unrecognised—and the anchor identity is scoped to the tested densities, where the differing high count is rare rather than combinatorially impossible.

Both results are, at root, a genotype–phenotype distinction. The rule table is the genotype: it fixes what is possible and can be read directly, which is why capability is *certifiable*. What a random trajectory develops is the phenotype: it depends on initial conditions and dynamics, so it can only be *estimated*, and it need not realise everything the genotype permits—a rule may carry a glider it never expresses (§3.3), just as two distinct genotypes may yield the same phenotype when the differing genes are never transcribed (the anchor variants above). The birth–survival asymmetry δ adds a smaller morphological note: translation requires a rule that distinguishes "ahead" from "behind", a broken front–back symmetry that in-place recurrence does not need—loosely, the asymmetry that a moving body plan requires and a radially symmetric one does not.

External corroboration, and the scope of the claim. Comparing against the literature's catalogue of complex life-like rules sharpens both what these results establish and what they do not. Of twenty rules recorded as Class IV or computationally universal in the standard references (Eppstein 2010, Adamatzky 2010), exactly one is totalistic (Replicator), and it is Class III rather than Class IV: *every* known Class IV rule in the family is non-totalistic. The $B = S$ exclusion is therefore corroborated externally, not only by our enumeration. Second, our behavioural screen, run without reference to the literature list, independently recovered four of these named rules—DryLife, EightLife, Pedestrian Life and LowDeath—and the subsequent table comparison correctly identified each as a Conway- or HighLife-family rule differing only at unvisited high neighbour counts. Third, and against these, the scoping must be stated plainly. Eleven of the twenty have $\Sigma_{3\times3} = 0$ and so could never enter our screen: Day & Night ($\lambda = 0.500$, $\mu = 0.565$), 2×2, Move and Life without Death are complex through *engineered* constructions and large-footprint propagation that a compact 3×3 enumeration cannot see. A vanishing compact-glider count therefore does not imply dynamical simplicity, and the $\Sigma = 0$ column of Figure 5 mixes inert rules with rules whose propagation is merely non-compact. The six *spontaneous* Class IV rules—the Conway family—do cluster where our coordinates predict ($\lambda \in [0.17, 0.22]$, $\mu \in [0.35, 0.46]$, $\delta \leq 0$). Our claim is accordingly the narrow one: among rules supporting a compact glider, spontaneous Game-of-Life–like ecology concentrates on Conway, HighLife and their behaviourally redundant variants. It is not the claim that these are the family's only complex rules—they demonstrably are not.

4.3 Limitations and open directions

The main enumeration is at 3×3 footprint, $T \leq 16$, exhaustive over all 262,144 rules. The 4×4 enumeration is also exhaustive over all rules (8,206 support a spaceship, 3.13%); the 5×5 enumeration is exhaustive over the 512 $B = S$ rules and sampled elsewhere. Three footprint observations follow. First, containment is perfect—all 424 $\Sigma_{3\times3} > 0$ rules retain $\Sigma_{4\times4} > 0$, with 7,782 further rules gliding only at 4×4—so the compact enumeration is a strict lower bound on propagation capability. Second, the storage axis is scale-invariant (within-slice Spearman $\rho(\#\mathrm{SL}, |S|) = 0.544$ at 3×3 and 0.556 at 4×4). Third, the $B = S$ exclusion is compact-specific but does not simply dissolve with scale: 36 of the 256 vacuum-stable $B = S$ rules admit a 4×4 spaceship and 40 a 5×5, yet those that combine a glider with static storage stay rare (4, then 8) and fall progressively further from Conway (still-life gap $6.3\times$ at 4×4, $9.2\times$ at 5×5), so the strict exclusion softens into a widening handicap. 6×6 and beyond are presently out of reach in both time and memory (the number of D_4 orbit-representative $n\times n$ seeds grows from $\sim 10^2$ at 3×3 to $\sim 10^4$, $\sim 10^6$, and $\sim 10^9$ at 4-, 5-, 6×6). The factorisation and the μ_{centre} mechanism are established empirically and argued heuristically; a proof that no $B = S$ rule admits a compact spaceship would convert the 3×3 exclusion into a theorem and is plausible from the centre-bit argument. The silent population was characterised across a 13-density, 25-seed sweep, but the underlying claim remains a measure, not an absolute: a basin of vanishing but nonzero weight cannot be excluded by sampling. A follow-up seeded with known spaceships rather than random ICs would map the basin structure directly, and the flood/freeze regime that gates expression—determined by the table yet not captured by a mean-field predictor—invites a sharper theory of which rules settle into the moderate-density band that propagation requires.

A The bit-flip duality and the dynamical sweep

The bit-flip duality σ (§2.4) is exact at the level of trajectories ($g(\neg x) = \neg f(x)$ for $g = \sigma(f)$) but is broken by sparse Bernoulli ICs: a rule and its dual visit different attractors from the same IC. Pairing every sample with its bit-flipped twin (the symmetric-IC ensemble of §2.6) restores the duality at the level of ensemble means; we verify a residual twin-gap at the seed-noise floor. The 512 σ-fixed points (e.g. Day & Night) are the self-dual rules. Beyond $\mathcal{G}^*_{\text{alt}}$ the sweep records per-frame compressibility, a basin-asymmetry index and the cell-flip rate; gauge-fixed by symmetric-IC averaging these co-locate σ-twins and separate canonical anchors into recognisable regimes, but—as §3.5 shows—do not predict $\Sigma > 0$ status and are reported only as regime descriptors, not capability detectors.

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